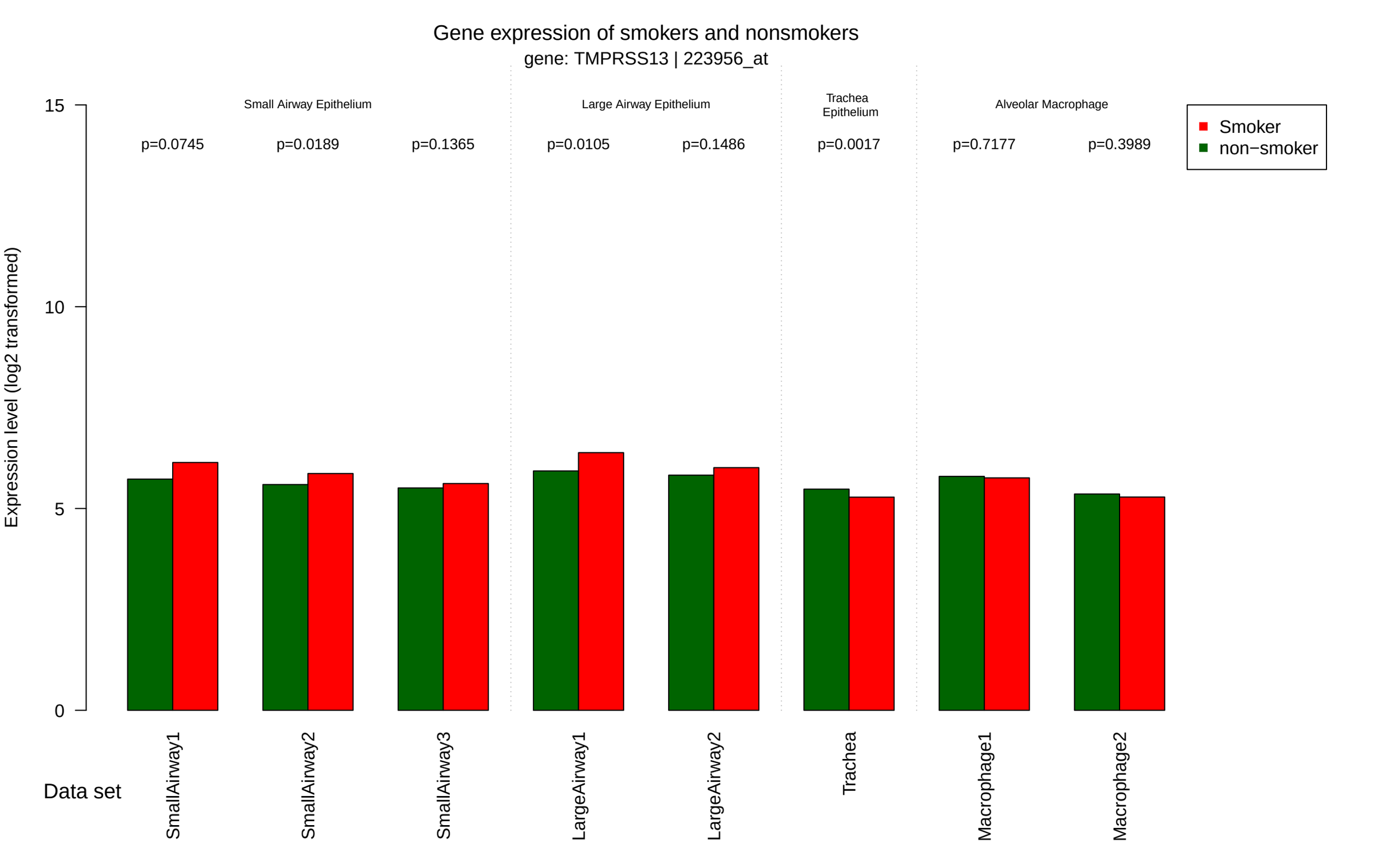
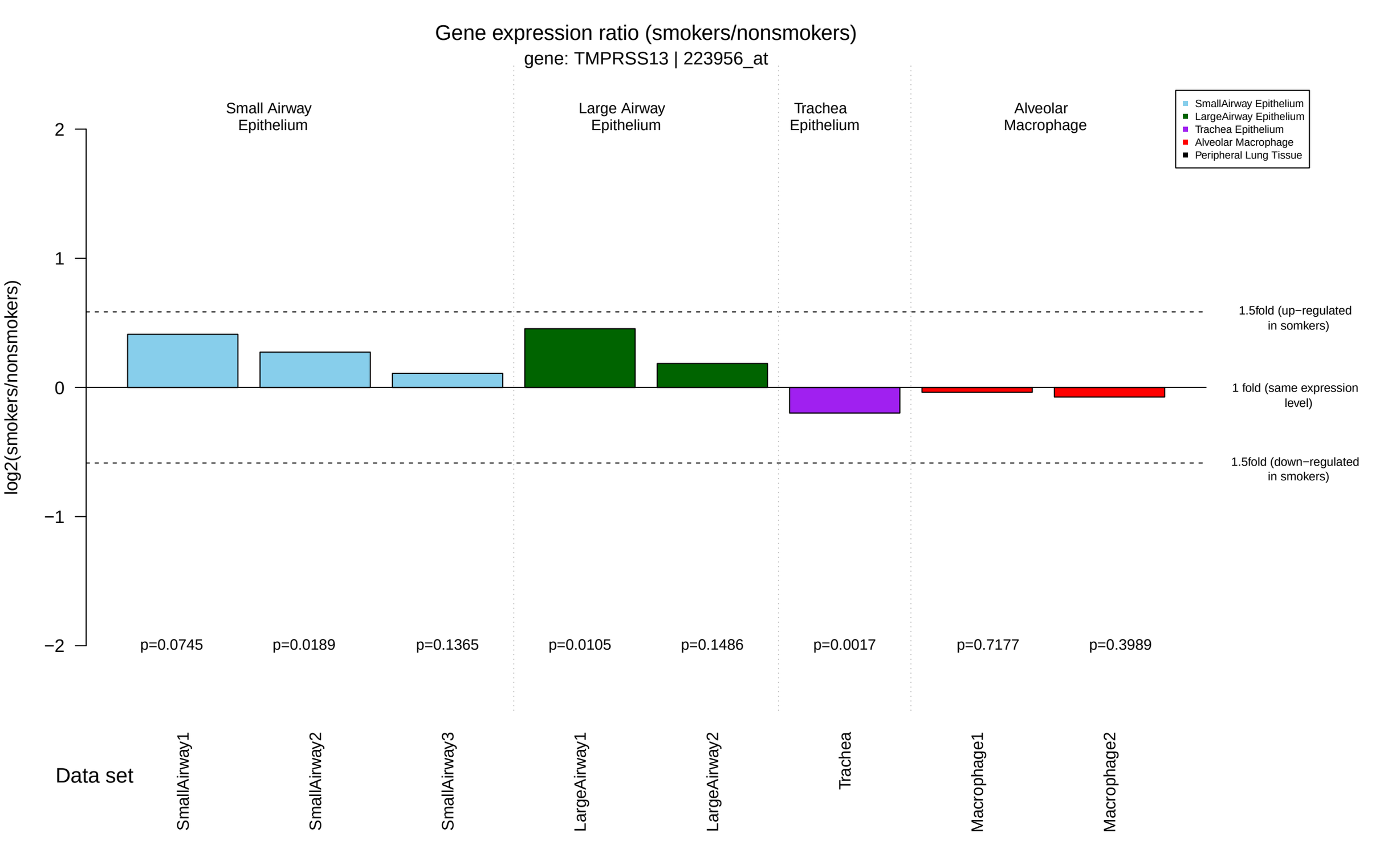
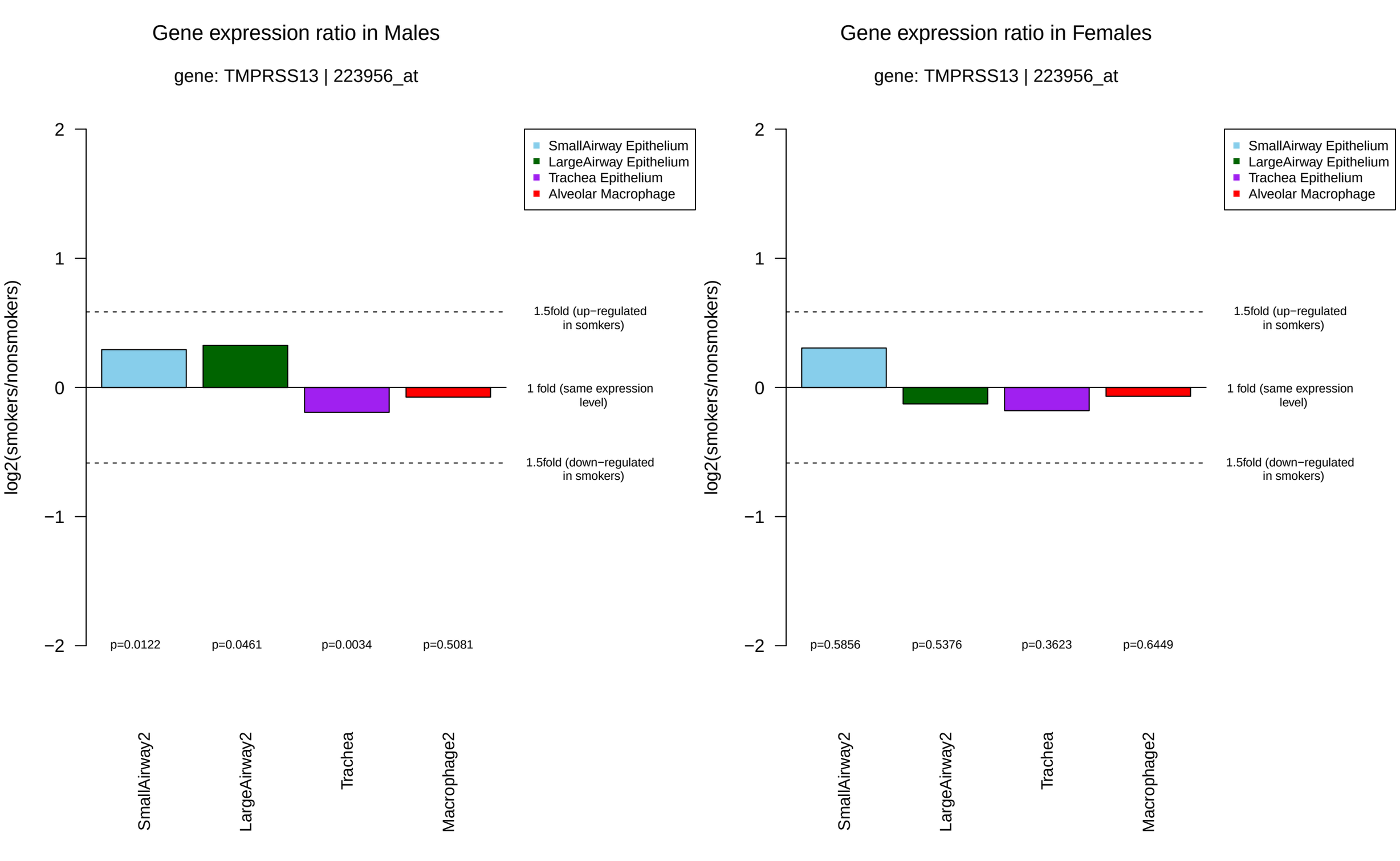
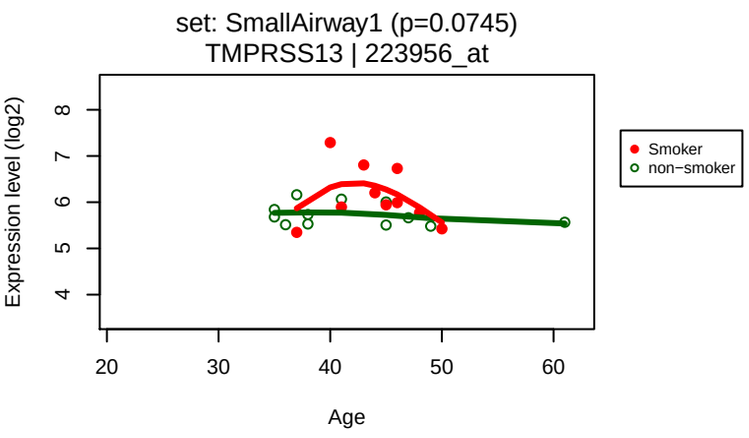
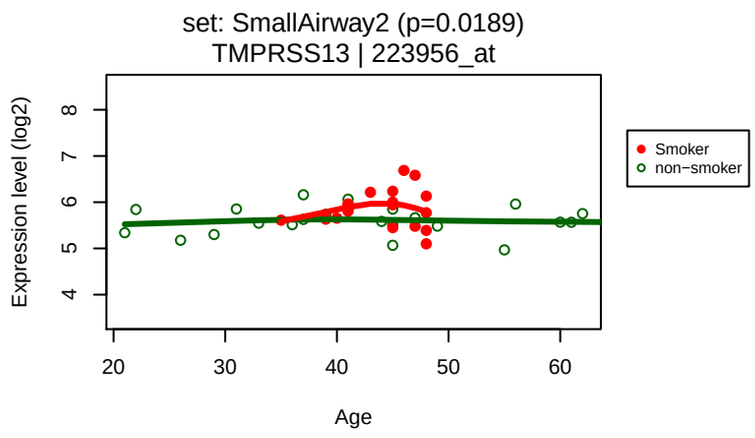
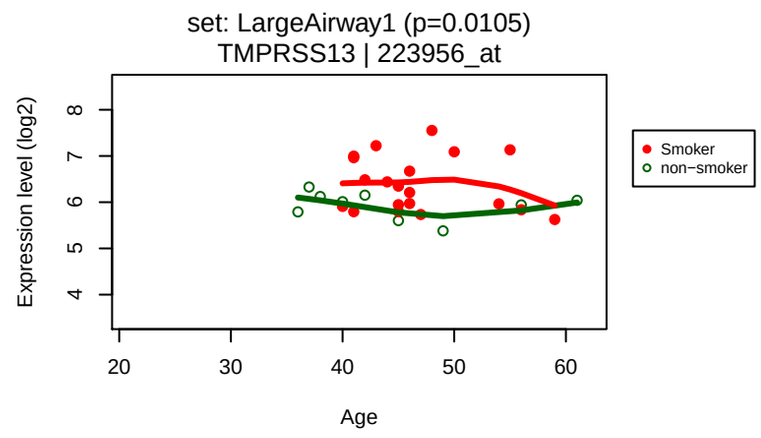

Gene expression of smokers and nonsmokers
gene: TMPRSS13 | 223956_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0745
p=0.0189
p=0.1365
p=0.0105
p=0.1486
p=0.0017
p=0.7177
p=0.3989
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

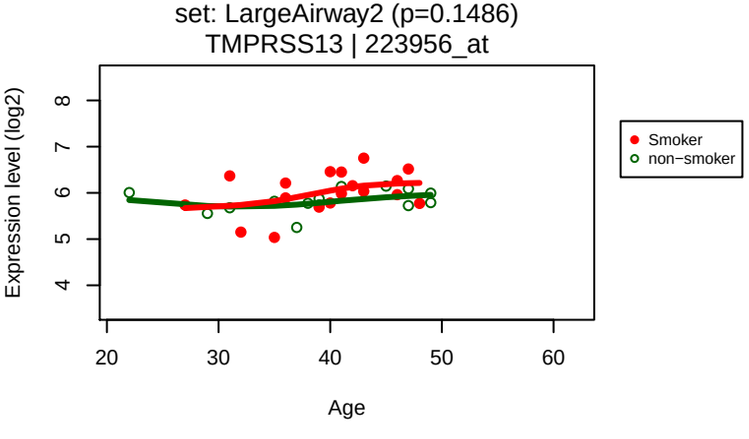

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS13 | 223956_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0745
p=0.0189
p=0.1365
p=0.0105
p=0.1486
p=0.0017
p=0.7177
p=0.3989
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

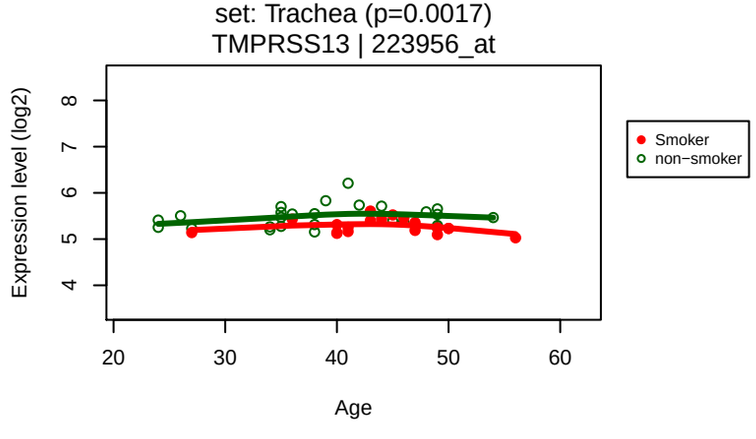

Gene expression ratio in Males
gene: TMPRSS13 | 223956_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.0461
p=0.0034
p=0.5081
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS13 | 223956_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5856
p=0.5376
p=0.3623
p=0.6449
SmallAirway2
LargeAirway2
Trachea
Macrophage2

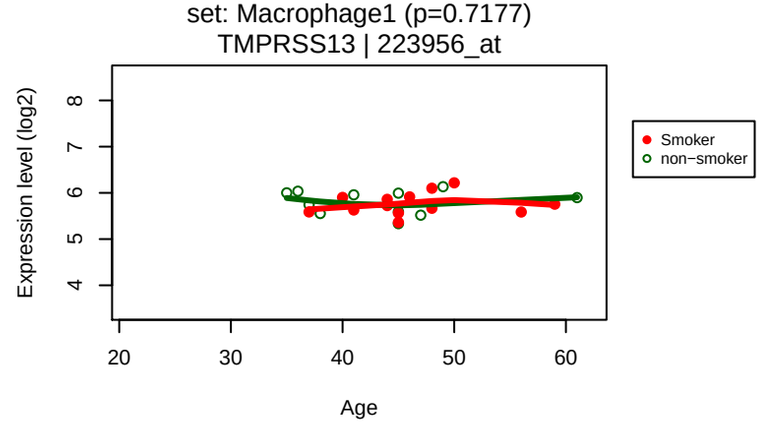

set: SmallAirway1 (p=0.0745)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker

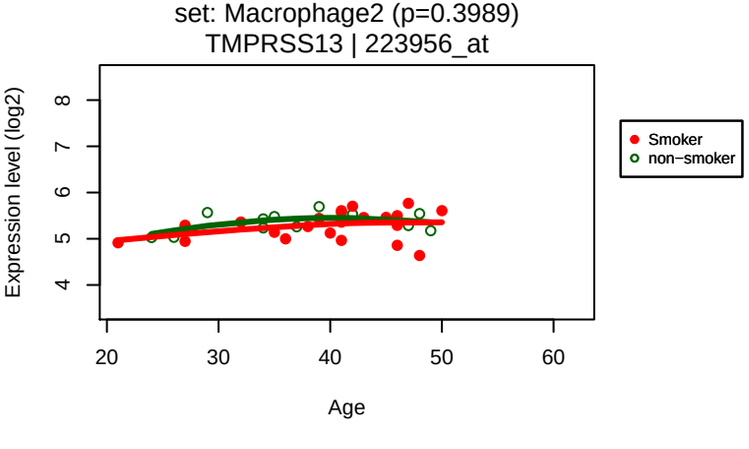

set: SmallAirway2 (p=0.0189)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker

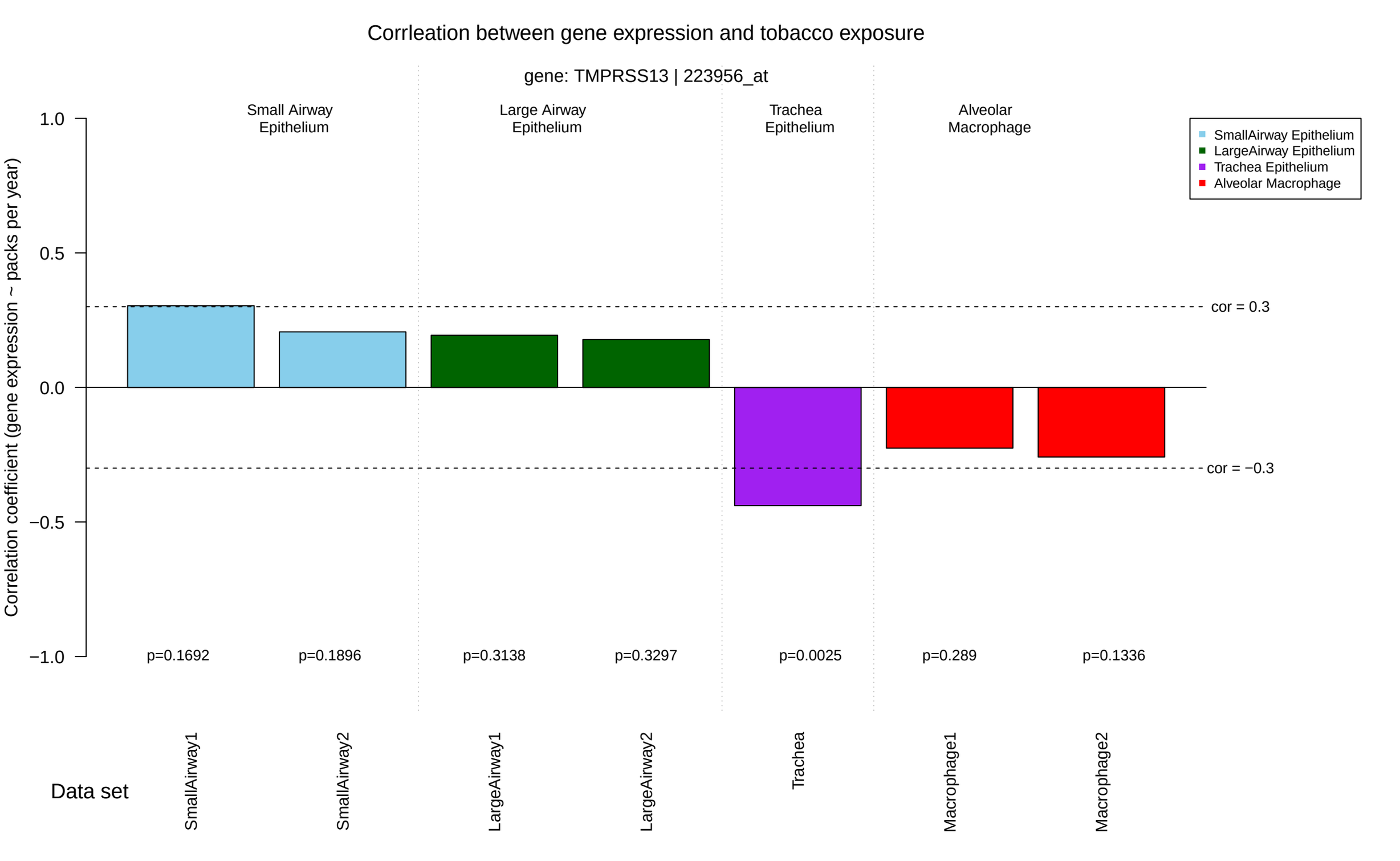

set: LargeAirway1 (p=0.0105)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1486)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0017)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7177)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3989)
TMPRSS13 | 223956_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS13 | 223956_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1692
p=0.1896
p=0.3138
p=0.3297
p=0.0025
p=0.289
p=0.1336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2